C S Abhinav

Bangalore, India — abhinavcs360@gmail.com — +91-9341238907 — linkedin.com/in/cs-abhinav — github.com/CSAbhinav3

Professional Summary

Data-oriented Computer Science student with hands-on experience analyzing real-world datasets to build, evaluate, and interpret predictive models. Skilled in exploratory data analysis, feature engineering, and defining meaningful evaluation metrics to avoid misleading conclusions. Strong foundation in machine learning, APIs, and analytics pipelines, with a focus on extracting actionable insights and supporting data-driven decision-making.

Technical Skills

Product & Analytics: Metrics definition, experimentation, data analysis, model evaluation
Programming: Python, Java, SQL
Backend / Systems: FastAPI, REST APIs, system integration
Data: Feature engineering, ETL, time-series analysis
Tools: Git, GitHub, Postman

Internship

AI-Based Business Profitability Estimator — Backend Intern, Dewdas Technology Apr – Jun 2025

- Contributed to a financial decision-support product estimating business profitability from structured inputs.
- Led the transition from Linear Regression to Random Forest based on performance and reliability considerations, improving test R^2 from 0.61 to 0.85.
- Defined evaluation metrics (MAE, R^2) to align model performance with business usability.
- Designed a FastAPI inference service with clear contracts to support frontend and product iteration.
- Deployed the service to cloud infrastructure (Render) for stakeholder testing.

Projects

Stock Return Prediction using ML & LSTM (Live Market Data)
Python, Pandas, NumPy, Scikit-learn, TensorFlow, LSTM

- Analyzed the feasibility of short-term stock return prediction using 5+ years of historical market data (OHLCV), comprising over 1,200 trading days per instrument.
- Implemented and evaluated multiple models including Linear Regression, Random Forest, and LSTM to study trade-offs between model complexity, prediction stability, and deployment feasibility.
- Assessed model performance using MAE, RMSE, and directional accuracy, achieving up to 58–62% directional accuracy on 1-day return prediction while highlighting limitations of absolute price forecasting.
- Performed exploratory data analysis and time-series visualization to identify volatility clustering, regime shifts, and model failure cases rather than relying on raw prediction outputs.
- Concluded that ML-based models provide limited edge for short-horizon returns under efficient market conditions, emphasizing risk-aware interpretation over naive profit claims.

IoT-Based Cold Chain Monitoring System
ESP32, ThingSpeak, Embedded Systems

- Built a 5-sensor IoT monitoring system (temperature, humidity, gas, vibration, GPS) using ESP32, sampling data at 10 s intervals for cold-chain surveillance.
- Implemented a threshold-based alert mechanism for temperature violations outside the 4–8 °C range, achieving alert response times under 3 s with zero false positives during testing.
- Streamed real-time telemetry to ThingSpeak with time-stamped logging, recording over 3,000 data points per experimental run with >99% data delivery reliability.
- Conducted 6+ hours of controlled cold-storage experiments using an insulated ice-box setup, successfully detecting and logging all induced compliance failures.
- Designed an end-to-end hardware-to-cloud pipeline with sub-5 s sensor-to-dashboard latency, enabling remote monitoring and traceability.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering
Christ University, Bengaluru
2023 – 2027
CGPA: 3.58 / 4.00 (as of January 2026)

Certifications

- Java Fundamentals — Oracle Academy
- Java Programming — Oracle Academy
- Introduction to MongoDB — MongoDB University